

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

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6 Conclusion and Future Work

This chapter synthesizes the dissertation, states the original contribution to knowledge, revisits the research questions, and outlines future work on cross-protocol validation, adversarial manipulation, hybrid methods, and live oracle integration.

6.1 Summary of findings

This dissertation formulated EndorseRank, a PageRank method on a latest-allowance endorsement graph that produces an on-chain reputation score on Arbitrum One, and evaluated it against Adaptive Weighted PageRank (AWP), the transfer-graph baseline of Do et al.^[16], on an identical matched wallet cohort ($n = 5,521$). Validation used five proxy families and GMX V2 perpetual outcomes over the observation window 2025-12-01 to 2026-05-31 (365,488 decoded `PositionDecrease` events, 14,727 EndorseRank edges, and 137,087 AWP edges).

The primary findings for the EndorseRank–AWP comparison are as follows.

On reputation structure (RQ1), EndorseRank and AWP yield partially overlapping yet construct-distinct orderings. Inter-method Kendall τ between the two score vectors is 0.316, within the weak-to-moderate band: correlation is present, but allowance and transfer graphs emphasize different wallet relationships. H1 is supported.

On social-method alignment (RQ2), construct-specific winners emerge. EndorseRank leads on allowance proxies ($\tau = 0.355$ versus AWP -0.030); AWP leads on transfer proxies ($\tau = 0.534$ versus EndorseRank 0.333) and on most social-outcome families. On trading-success proxies specifically, AWP exceeds EndorseRank ($\tau = 0.179$ versus 0.096). H2 is supported in its construct-specific form: neither social method universally dominates.

On computational efficiency (RQ3), EndorseRank requires less PageRank wall-clock time than

AWP on the matched cohort and on the expanded wallet pool. At $n = 5,521$, runtimes are 2.105 s versus 18.711 s ($\approx 8.9\times$). Pool scaling preserves the advantage at $n = 10,000$ ($\approx 10.4\times$) and at $N = 31,612$ ($\approx 9.4\times$). Robustness checks on damping and top-token subgraphs do not overturn the efficiency–alignment pattern. H3 is supported (Claim C1); together with alignment results, this feeds Claim C4.

On validation frames (RQ4), transfer proxies yield stronger rank alignment than trading-success proxies for both social methods. Outcome-based proxies show weak alignment overall, indicating that social-method rankings align more strongly with transfer centrality than with perpetual trading profitability on this sample.

As a supplementary finding (SRQ1), GF-PR achieves the highest trading-success mean τ (0.583) and inverse-risk mean τ (0.376) by construction, illustrating construct overlap when ranking on outcome flow. GF-PR bounds trading-outcome proxy interpretation for social methods and is not a deployable social reputation score (Claim SC1).

Collectively, Claims C1–C4 establish an efficiency–alignment trade-off: EndorseRank offers substantially lower PageRank cost and stronger allowance-construct validity, while AWP offers stronger transfer and social-outcome alignment. The appropriate choice depends on whether operators prioritize authorization semantics and refresh latency or transfer-hub detection and outcome co-movement.

6.2 Original contribution to knowledge

This dissertation contributes to on-chain reputation research and DeFi risk assessment in four respects.

- **EndorseRank:** An allowance-based endorsement graph model. By treating latest ERC-20 approve and EIP-2612 permit authorizations as directed endorsement edges ^{[18],[19]}, it extends PageRank’s citation metaphor ^{[8],[49]} from transfer graphs ^[16] to explicit spending delegation. The model eliminates temporal decay through allowance override semantics and yields

a sparse graph (14,727 edges on the matched cohort) suitable for frequent recomputation.

- **Five-proxy domain alignment evaluation:** Compares EndorseRank and AWP on leveraged DeFi activity under a shared cohort and window. Building on construct-validity reasoning^[14] and rank correlation measures^{[36],[54]}, the evaluation separates transfer, allowance, inverse-risk, Sybil-adjusted stability, and trading-success families. The result is a construct-specific map of strengths rather than a single global ranking—a methodological template for future on-chain score comparisons.
- **Reproducible open-source evaluation pipeline:** Links extraction, graph construction, scoring, proxy computation, and LaTeX-ready tables to a machine-readable summary (Appendix 8.1). Auditability of decentralized credit-related scores is a recognized concern^[48]; the pipeline makes the empirical claims independently verifiable.
- **GF-PR ceiling analysis:** Clarifies construct overlap on trading-outcome proxies (SRQ1). By ranking directly on GMX PnL flow, GF-PR sets an outcome-native upper bound ($\tau = 0.583$ on trading success; $\tau = 0.376$ on inverse risk) against which social-method alignment can be interpreted without mistaking shared inputs for social validity. Supplementary contribution only.

Relative to prior graph-reputation and DeFi credit work^{[16],[27],[29],[35],[39],[44]}, the original knowledge contribution is the demonstration that allowance-graph PageRank is a distinct, efficient, and allowance-valid social reputation construct on Arbitrum One leveraged trading data—not a drop-in approximation of transfer-graph AWP, and not a substitute for collateralization^{[26],[28],[53],[59]}.

Authorization graphs and payment graphs are therefore not interchangeable reputation substrates. They differ in sparsity (14,727 versus 137,087 edges on the matched cohort), runtime (2.105 s versus 18.711 s), and alignment profiles (allowance $\tau = 0.355$ for EndorseRank; transfer $\tau = 0.534$ for AWP), and they should be validated against proxies that match their intended semantics^[14].

6.3 Future work

Several extensions follow from the limitations in Chapter 5 and the scope of this study.

- **Cross-protocol and cross-chain validation:** Replication on other perpetual markets, lending protocols, and Layer-2 or Layer-1 networks would test whether the allowance–transfer trade-off generalizes beyond Arbitrum One and GMX V2^{[53],[59]}. Where lending default or liquidation labels are denser, credit-ground-truth validation can complement the trading-success frame used here^{[26],[44]}. Cross-chain GMX deployments and alternative L2s offer natural replication sites with comparable event schemas.
- **Adversarial evaluation:** The Sybil cost model in Chapter 5 is analytical. Empirical red-teaming—including wash approvals, coordinated spender farms, and strategic revocation—would quantify realized attack cost-effectiveness under live fee markets and detection heuristics^[17]. Sybil clustering on G_{approve} and comparison with transfer-graph manipulation costs are natural next experiments. Such work would also stress-test whether Sybil-adjusted stability proxies remain informative under active adversaries.
- **Hybrid social and outcome-aware methods:** Operators may combine endorsement-graph features with transfer-graph features, or with DeFi-informed representation learning^[39], while preserving the auditability emphasized by Packin and Lev-Aretz^[48]. Outcome-native diagnostics in the spirit of GF-PR can remain on a supplementary track for bounding trading-outcome proxy evaluation without being deployed as social scores. Hybrid designs should report construct-specific alignment rather than a single aggregate metric, following the five-proxy template of this dissertation.
- **Live oracles and governance tooling:** Integration with risk oracles, liquidation prioritization, and governance dashboards would test whether EndorseRank’s runtime profile (2.105 s on the matched cohort; order-of-magnitude gains on the expanded pool) sustains production refresh cadences. Careful interface design is required so that ordinal graph scores are not over-interpreted as consumer credit ratings. Human-in-the-loop safeguards remain appropri-

ate wherever scores gate access to financial services.

- **Collateral-differentiation experiments:** Section 5.4.4 argues that protocols can capture value through fee tiers, utilization, and monitoring cost without treating reputation as a collateral substitute. Controlled A/B deployments that modestly differentiate collateral or limit-increase cadence by EndorseRank or AWP bands—with explicit stop-loss rules and disclosure of weak trading-success alignment—would test whether such policies increase volume or utilization without elevating default or liquidation rates. Until such evidence exists, collateral reduction remains an adoption hypothesis rather than an empirical claim of this dissertation.
- **Adoption pathway and reputation multiples:** Longitudinal studies of protocols that publish recomputable social ranks would clarify whether a shared ordinal vocabulary spreads across venues in the manner of valuation multiples (Section 5.4.5). Metrics of interest include cross-protocol score correlation, fee-tier uptake, monitoring-cost deltas, and Sybil-filter efficacy under live reward programs.
- **Archived baselines and richer authorization semantics:** Extended baselines archived in Appendix 8.1 can be revisited when additional social-graph semantics are needed. Richer authorization events—including protocol-specific delegation logs where available—may refine endorsement edge definitions beyond ERC-20 allowances alone. Longer observation windows and additional market regimes would further stress-test stability of the reported τ profiles.

6.4 Closing reflection

On-chain reputation will not dissolve the need for collateral in adversarial, pseudonymous markets [26],[53]. It can, however, make behavioral structure legible. This dissertation shows that legibility depends on which edges we choose to read. Transfers tell a story about economic flow; allowances tell a story about delegated authority. PageRank can narrate either story, but not both at once with the same ranking. EndorseRank commits to the authorization narrative, documents its costs and

benefits, and refuses to confuse outcome-native ceilings with social reputation evidence.

The practical upshot is modest and actionable. If a protocol or researcher needs a fast, auditable score of who is trusted to spend, EndorseRank is a credible instrument on Arbitrum One data of the kind studied here. If the goal is to find transfer hubs or to track activity-correlated outcomes, AWP remains stronger among social methods. If the goal is to predict trading profits from social graphs alone, neither method is sufficient—and GF-PR warns against evaluations that bake the answer into the graph.

This dissertation reports primary empirical claims C1–C4 addressing RQ1–RQ4 and supplementary claim SC1 (SRQ1) on a matched Arbitrum One wallet cohort ($n = 5,521$). EndorseRank stands as an efficient, allowance-specific alternative to transfer-based AWP for on-chain reputation under pseudonymity: distinct in ranking geometry (inter-method $\tau = 0.316$), aligned with its primary construct (allowance $\tau = 0.355$), and approximately $8.9\times$ faster to compute on the studied sample, with sustained advantages on the expanded wallet pool ($\approx 10.4\times$ at $n = 10,000$; $\approx 9.4\times$ at $N = 31,612$). Those properties motivate continued research on authorization-aware trust signals in decentralized markets.